

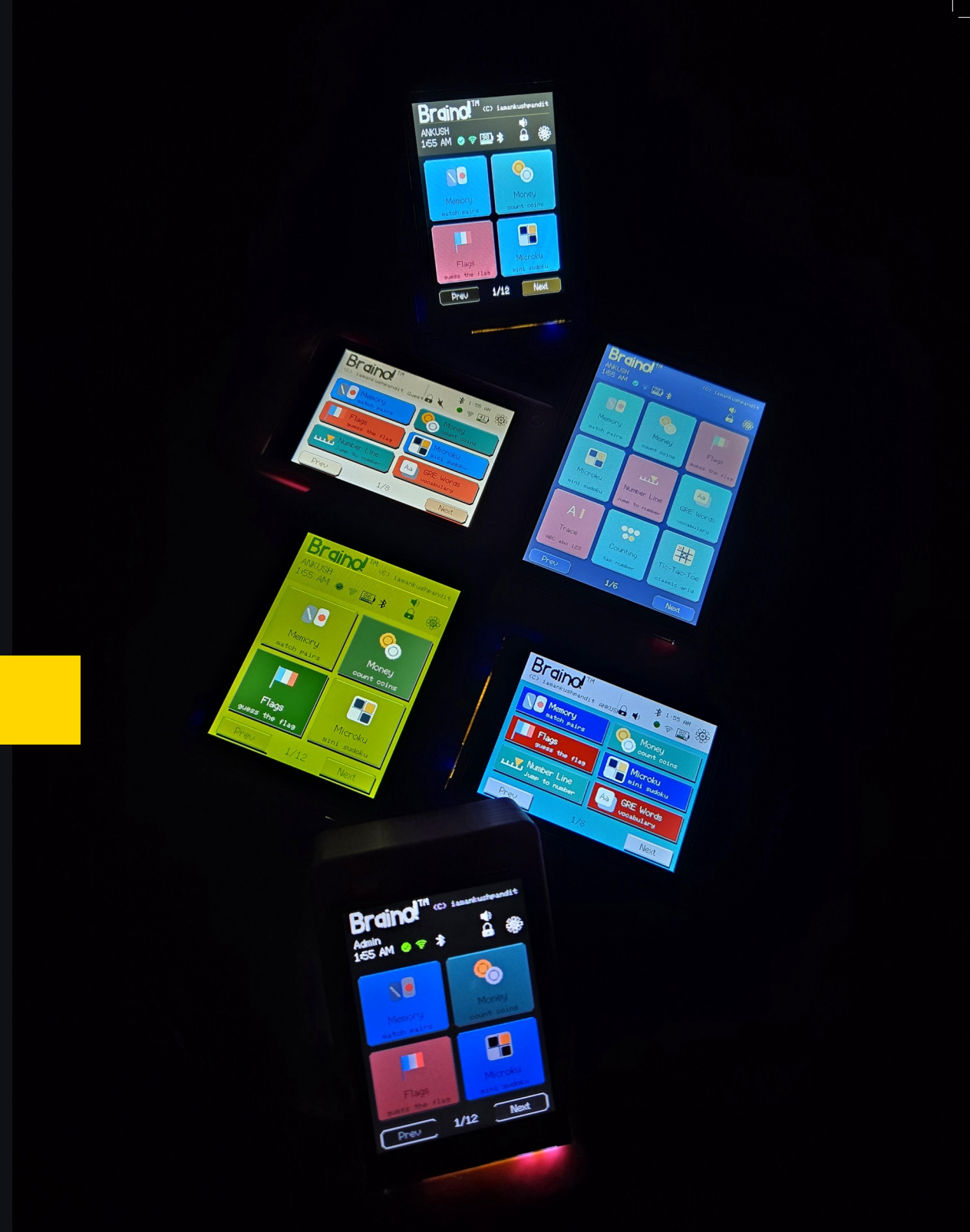
Braino!

OPEN-SOURCE LEARNING CONSOLE FIRMWARE

4 MB.

That is smaller than one photo. It runs an entire learning console.

Games that teach handwriting, maths, flags, the periodic table and more — on an ESP32 CYD touchscreen board, with no cloud, no accounts and no ads. Everything fits in the 4 MB of flash on the board.



In a world drowning in gigabytes and cloud storage, Braino fits an entire learning platform within four megabytes.

IT IS FIRMWARE

Not a gadget to buy. Braino runs on a CYD — a "Cheap Yellow Display": an ESP32 already soldered to a touchscreen. Search ESP32 CYD on AliExpress or Amazon.

FLASH IT FROM CHROME

Plug in USB-C, open the site in Chrome or Edge, pick your board, press Flash. About a minute. No toolchain, no SD card, no soldering.

OPEN SOURCE. FREE.

GPL-3.0. Firmware and 3D-printable cases live in one repo. No version to buy, no subscription, nothing phones home.

WHICH BOARD

2.8" resistive or capacitive
3.2" resistive
4" resistive

one firmware — pick the board when you flash

PLAYERS, EVERY AGE

First letters to the periodic table to brain training. Spaced repetition and adaptive difficulty, per player. Learning that looks like play.

ADMINISTRATORS

You curate which games each player sees — not every game for everyone. Up to five profiles plus Guest. No ads, no accounts, no feed, no telemetry.

PLAY TOGETHER

Chess, Sea Battle and Ludo across consoles over Bluetooth Low Energy. A beacon, not pairing: consoles find each other, and the beacon is on only when an administrator turns it on. Dice are never sent — each console rolls for itself.

ESP32 MAKERS

Your CYD is bored. No OS, no SD card, no cloud — just 4 MB of flash doing all of it. Five boards supported, six themes, self-playing Pong screensaver with the case LED following the rally.

3D PRINTING COMMUNITY

The case is in the repo. Prints without supports, no screws, lid snaps on. Room inside for a 3000 mAh battery and a speaker. Pink is the house colour — print yours in anything.

GAME DEVELOPERS

Add a game. The launcher, profiles, scoring, mastery tracking and spaced repetition already exist — you write the activity, Braino does the rest.

BOARD PORTERS & HARDWARE PARTNERS

A new board is one header file plus one platformio.ini section; docs/PORTING.md is the checklist. CYD is only the demo board — Braino is ready for a purpose-built one.

CASE DESIGNERS

More boards need a case. Bring CAD skills: the 2.8, 3.2 and 4-inch boards all want a good enclosure, in speaker and no-speaker variants.

BUILT WITH AI. ON PURPOSE.

The privacy contract is written into the repo — no telemetry, no accounts, nothing leaves the board — and CI checks every change against it, human or AI. AI did not make it less safe. It made it provably safe.



Three consoles, one table, no phone



Inside: battery, speaker, board. That is the whole assembly.



The lid: speaker grille, embossed mark, BOOT and RESET slots



Star it on GitHub.

github.com/iamankushpandit/Gume

Flash it from your browser: iamankushpandit.github.io/Gume

One star helps it get found.

Free • Open source • GPL-3.0

No ads • No accounts • No cloud